

★ Clock

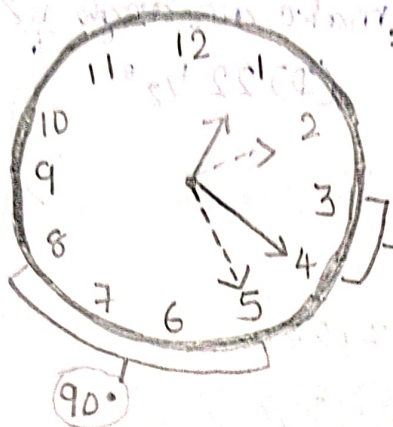
Hour Hand Minute Hand

⇒ Hands. [Minimum Hands (H.H) & (M.H)]

⇒ Movement [Clockwise (same direction)]

⇒ Relative Speed [same direction]
Subtract (-)

⇒ Time → Angle?
⇒ Angle → Time?
⇒ Gain OR Lose?
[Patterns].



⇒ 360°
12 parts ⇒ $\frac{360}{12} \Rightarrow 30^\circ$

⇒ Minute Hand (M.H)
↳ 4 to 5 [5 mins]

If dist. = 30°

∴ Speed = $\frac{30^\circ}{5} = 6^\circ/\text{min.}$ M.H

⇒ Relative Speed

Subtract M.H, H.H.

$6^\circ/\text{min} - 0.5^\circ/\text{min}$

⇒ $5.5^\circ/\text{min}$

⇒ Hour Hand (H.H)

↳ 1 to 2 [60 mins]

If dist. = 30°

∴ Speed = $\frac{30^\circ}{60} = \frac{1}{2}^\circ/\text{min}$ H.H.
OR
 $0.5^\circ/\text{min}$

★ Question asked like: [Patterns]

(i) Time → Angle?

(ii) Angle → Time?

(iii) Gain OR Lose.

• First Pattern [Random Time Given]



⇒ 2:00 (60°)

⇒ 7:00 (210°)

⇒ 4:00 (120°)

⇒ 8:00 (240°)

Q ⇒ 4:45

↳ 4 hrs → 120°

45 mins × 5.5°

⇒ 247.5°

Ans ⇒ 127.5°

Taking difference

247.5

-120

127.5

Bigger-Smaller

⇒ If timing given

by rounded of time.

then use time × 30°

Q ⇒ 5:30

↳ 5 hrs → 150°

30 mins × 5.5° ⇒ 165

Ans ⇒ 15°

Q ⇒ 7:23

↳ 7 hrs → 210°

23 mins × 5.5°

⇒ 126.5°

Ans ⇒ 83.5°

210°

126.5

83.5

$$Q \Rightarrow 3:15$$

$$\hookrightarrow 3 \times 30 \Rightarrow 90^\circ$$

$$15 \times 5.5 \Rightarrow 82.5$$

$$\Rightarrow 90 - 82.5 = 7.5^\circ$$

$$Q \Rightarrow 4:40$$

$$\hookrightarrow 4 \times 30 \Rightarrow 120^\circ$$

$$40 \times 5.5 \Rightarrow 220$$

$$\Rightarrow 220 - 120$$

$$\Rightarrow 100^\circ$$

clock

Que $\Rightarrow$ At 9:45, the two hands of a clock make an angle of?
 (a) 60° (b) 45° (c) $33\frac{1}{3}^\circ$ (d) $22\frac{1}{2}^\circ$

$$\Rightarrow 9 \times 30^\circ \Rightarrow 270^\circ$$

$$45 \times 5.5 \Rightarrow 247.5$$

$$\text{Ans} \Rightarrow 270 - 247.5$$

$$\Rightarrow 22.5$$

If...



$$\Rightarrow 22.5^\circ \checkmark$$

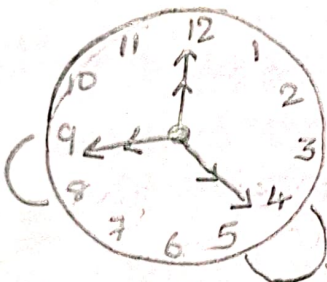
$$\Rightarrow 360 - 22.5^\circ$$

$$\Rightarrow 337.5^\circ \checkmark$$

} Both Ang Correct.

• Second Pattern [Angle $\rightarrow$ Time(?)]

$\Rightarrow 0^\circ$ OR Coincide: H.H and M.H coincide each other.



$\Rightarrow$ Every one hour minute hour & hour hand coincide. But...

(Note: If you take hour between 11 and 12 OR 12 and 1;

$\Rightarrow$ Both hands coincide at 12:00 only.

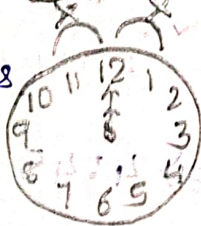
$\Rightarrow$ Either 12 and 1 or 11 and 12 takes one coincide.

Exception

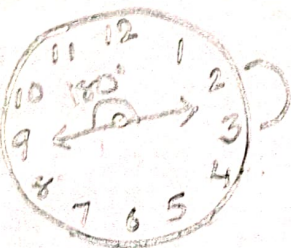
$$\therefore 12 \text{ hrs} \Rightarrow 11 \text{ times}$$

$$\therefore 24 \text{ hrs} \Rightarrow 22 \text{ times}$$

Coincide



$\Rightarrow 180^\circ$ OR Opposite: H.H and M.H opposite each other.



$\Rightarrow$ Every one hour minute hour & hour hand opposite to each other. But

(Note: If you take hour between 5 and 6 OR 6 and 7;

$\Rightarrow$ Both hands having opposite hand at 6:00 only

$\Rightarrow$ Either 5 and 6 or 6 and 7 take one opposite,

Exception

$$\therefore 12 \text{ hrs} \Rightarrow 11 \text{ times}$$

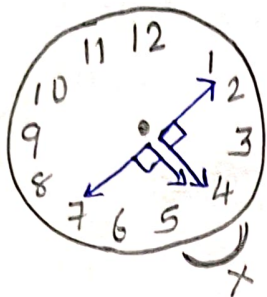
$$\therefore 22 \text{ hrs} \Rightarrow 22 \text{ times}$$

Opposite.



⇒ Except 0° and $180^\circ \Rightarrow 1 \text{ hour} \Rightarrow 2 \text{ Times}$.

⇒ 90° or Right angle := [2 Times]

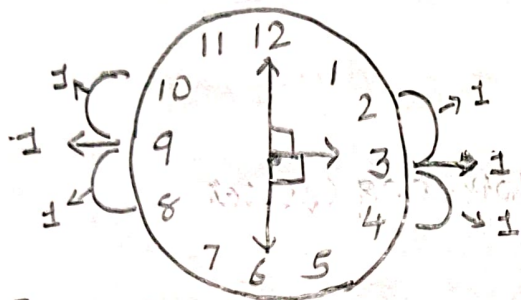


⇒ In between 4 and 5, Two times Right angle occurs

⇒ **NOTE:** In between 2 and 3; it comes for 2 times. &

⇒ In between 3 and 4; it comes for 2 times....

⇒ That means; between 2 and 4 it comes for 4 times but exception is between 2 and 4 it comes for only 3 times.



⇒ Either 2 and 3 having both angles and 3 and 4 having 1 OR 3 and 4 having both angles and 2 and 3 having 1.

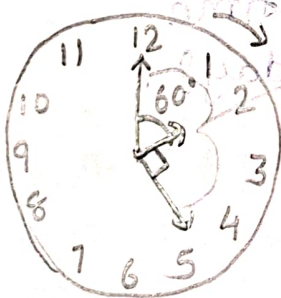
⇒ Same for 8, 9, 10 Reflexive angle.

Exception

12 hours $\Rightarrow$ 22 Times

24 hours $\Rightarrow$ 44 Times.

Que In between 2'0 clock and 3'0 clock at what time the hands of clock form 90° ?



⇒ after covering 60° then after move towards $90^\circ \Rightarrow 60^\circ + 90^\circ = 150^\circ$

⇒ Because before 2'0 it cannot be count the degree so we have to move forward.

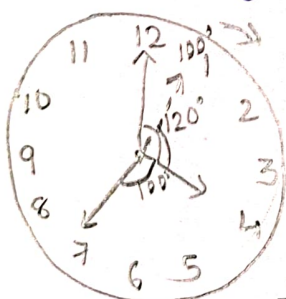
⇒ Dist $\Rightarrow 150^\circ$; Speed $\Rightarrow 5.5$

$$\text{Time} = \frac{\text{dist}}{\text{speed}} = \frac{150}{5.5} = 27 \frac{3}{11} \text{ minute.}$$

Between 2 and 3.....
 $90^\circ \rightarrow$

2 hrs : $27 \frac{3}{11} \text{ min}$

Que In between 4'0 clock and 5'0 clock at what time the hands of clock form 100° ?



⇒ After moving towards M.M to H.M it is possible that 100° comes..

⇒ After moving 20° towards (100° comes).

$$\frac{20}{5.5} = 3 \frac{7}{11}$$

⇒ 4 hrs : $3 \frac{7}{11} \text{ mins}$ (100°)

(I)

$$120^\circ + 100^\circ \Rightarrow 220$$

$$\Rightarrow \frac{220}{5.5} = 40$$

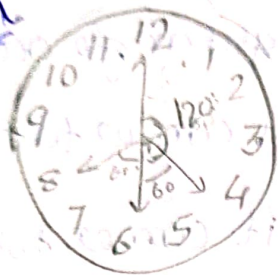
⇒ 4 hrs : 40 mins (100°)

(II)

Que

In between 6'o clock and 7'o clock at what time the hands of clock form 60° ?

Sol



$$(I) \frac{120}{5.5} \Rightarrow \frac{240}{11}$$

$$\Rightarrow 21 \frac{9}{11}$$

$$\Rightarrow \text{6 hrs : } 21 \frac{9}{11} \text{ mins}$$

$$(II) \frac{240}{15.5} \Rightarrow \frac{480}{11}$$

$$\Rightarrow 43 \frac{7}{11}$$

$$\Rightarrow \text{6 hrs : } 43 \frac{7}{11} \text{ mins}$$

Que In between 1'o clock and 2'o clock at what time the hands of clock form 100° ?

Sol



$$(I) 30^\circ + 100^\circ \Rightarrow 130^\circ$$

$$\Rightarrow \frac{130}{5.5} \Rightarrow \frac{260}{11} \Rightarrow 23 \frac{7}{11}$$

$$\Rightarrow \text{1 hr : } 23 \frac{7}{11} \text{ min}$$

(II) Take Reflex Angle

$$\Rightarrow 360^\circ - 100^\circ \Rightarrow 260^\circ$$

$$260^\circ + 30^\circ$$

$$\Rightarrow \frac{290}{5.5} = \frac{580}{11} = 52 \frac{8}{11}$$

$$\Rightarrow \text{1 hrs : } 52 \frac{8}{11} \text{ min}$$

• Third Pattern [Gain OR Lose]

Que A clock which gains 5 minutes in every one hour was set correct at 5 a.m. What would be time shown by that clock at 1 pm the same day?

Sol

5 a.m. $\rightarrow$ 1 pm [8 hours]

$$8 \times 5 \Rightarrow 40 \text{ mins}$$

1 pm $\rightarrow$ 1:40 mins

Que A clock looses 10 mins every one hour was set correct at 4 a.m. What would be the time shown by the clock at 4 pm the same day?

4 a.m. $\rightarrow$ 4 pm [12 hrs]

$$12 \times 10 \Rightarrow 120 \text{ mins}$$

4 pm $\rightarrow$ 2 pm

$$4 \text{ pm} - 120 \text{ mins} \Rightarrow 2 \text{ pm}$$

(2 hrs)

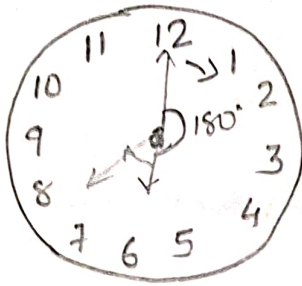
Que At what time between 6 AM and 7 AM will the minute hand and hour hand of a clock make angle closest to 60° ?

(A) 6:22 AM

(B) 6:27 AM

(C) 6:38 AM

(D) 6:45 AM



$$\Rightarrow \frac{120^\circ}{5.5} \Rightarrow \frac{240}{11} \Rightarrow 21 \frac{9}{11} \quad \Rightarrow 6 \text{ hrs} : 21 \frac{9}{11} \quad \text{I}$$

$$\Rightarrow \frac{240}{5.5} \Rightarrow \frac{480}{11} = 43 \frac{7}{11} \quad \Rightarrow 6 \text{ hrs} : 43 \frac{7}{11} \text{ min} \quad \text{II}$$

- Mechanical Engg. [GATE 2019]
- 1 Mark.

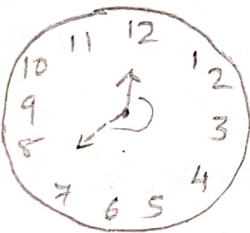
Que A worker noticed that the hour hand on the factory clock had moved by 225° during her stay at the factory. For how long did she stay in the factory.

(A) 3.75 hrs

(B) 4 hrs 15 min

(C) 8.5 hrs

(D) 7.5 hrs



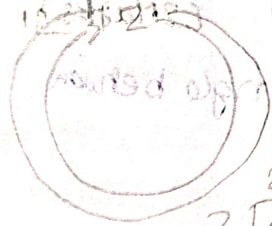
Hour hand $\rightarrow \frac{1}{2}^\circ/\text{minute}$.

$$\begin{aligned} 0.5^\circ &\Rightarrow 1 \text{ min} \\ 1^\circ &\Rightarrow 2 \text{ min} \\ 225^\circ &\Rightarrow (?) \end{aligned}$$

$$\begin{array}{r} 450 \text{ min} \\ 7.5 \\ 60 \overline{) 450} \\ \underline{420} \\ 30.0 \end{array}$$

③

12:00 to 12:25 min



23 hrs

1 hr $\rightarrow 600 \text{ sec}$

47 hrs

23.5

2 | 47

23.5

23.5

23.5

23.5

23.5

23.5

23.5

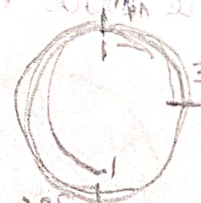
23.5

23.5

23.5

23.5

23.5

③ 0 min $\rightarrow 10 \text{ sec}$ 60 min $\rightarrow 600 \text{ sec}$ 1 hr $\rightarrow 600 \text{ sec}$ 2 hrs $\rightarrow 1200 \text{ sec}$ 3 hrs $\rightarrow 1800 \text{ sec}$ 4 hrs $\rightarrow 2400 \text{ sec}$ 5 hrs $\rightarrow 3000 \text{ sec}$ 6 hrs $\rightarrow 3600 \text{ sec}$ 7 hrs $\rightarrow 4200 \text{ sec}$ 8 hrs $\rightarrow 4800 \text{ sec}$ 9 hrs $\rightarrow 5400 \text{ sec}$ 10 hrs $\rightarrow 6000 \text{ sec}$ 11 hrs $\rightarrow 6600 \text{ sec}$ 12 hrs $\rightarrow 7200 \text{ sec}$ 13 hrs $\rightarrow 7800 \text{ sec}$ 14 hrs $\rightarrow 8400 \text{ sec}$ 15 hrs $\rightarrow 9000 \text{ sec}$ 16 hrs $\rightarrow 9600 \text{ sec}$ 17 hrs $\rightarrow 10200 \text{ sec}$ ④ 12 min $\rightarrow 720 \text{ sec}$ 1 hr $\rightarrow 3600 \text{ sec}$ 2 hrs $\rightarrow 7200 \text{ sec}$ 3 hrs $\rightarrow 10800 \text{ sec}$ 4 hrs $\rightarrow 14400 \text{ sec}$ 5 hrs $\rightarrow 18000 \text{ sec}$ 6 hrs $\rightarrow 21600 \text{ sec}$ 7 hrs $\rightarrow 25200 \text{ sec}$ 8 hrs $\rightarrow 28800 \text{ sec}$ 9 hrs $\rightarrow 32400 \text{ sec}$ 10 hrs $\rightarrow 36000 \text{ sec}$ 11 hrs $\rightarrow 39600 \text{ sec}$ 12 hrs $\rightarrow 43200 \text{ sec}$ 13 hrs $\rightarrow 46800 \text{ sec}$ 14 hrs $\rightarrow 50400 \text{ sec}$ 15 hrs $\rightarrow 54000 \text{ sec}$ 16 hrs $\rightarrow 57600 \text{ sec}$ 17 hrs $\rightarrow 61200 \text{ sec}$ 

1 hr $\rightarrow 25 \text{ min}$ 21 hr
21 hr $\rightarrow 27 \frac{1}{2} \text{ min}$
8.75 hrs

21 hr $\rightarrow 525 \text{ min}$ 1260
 $\Rightarrow 735$

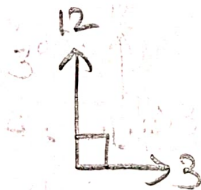
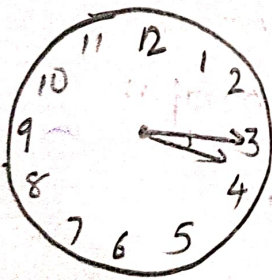
12.25

Assignment Question

-Electronics Engg. - GATE 2020
[2 Marks]

Que. It is quarter past three in your watch. The angle between the hour hand and the minute hand is.

Sol.



- 15 min: (A) 0°
(B) 15°
(C) 7.5°
(D) 22.5°

$$15 \times 5.5 \Rightarrow 82.5$$

$$\begin{array}{r} 90^\circ \\ - 82.5 \\ \hline 7.5 \end{array}$$

$$\begin{array}{r} 11 \overline{) 110} \\ 11 \\ \hline 00 \\ 00 \\ \hline 00 \\ 00 \\ \hline 00 \end{array}$$